their authority to a third-party app, restricting how much access they grant using scopes.

Whereas scopes are used for delegation, permissions may be used for either mandatory or discretionary access. File permissions in UNIX and most other popular operating systems can be set by the owner of the file to grant access to other users and so implement DAC. In contrast, some operating systems used by the military and governments have mandatory access controls that prevent somebody with only SECRET clearance from reading TOP SECRET documents, for example, regardless of whether the owner of the file wants to grant them access.[3] Methods for organizing and enforcing permissions for MAC are covered in chapter 8.

In Natter, permissions are granted by the owner of a social space when adding a new member, but then cannot be further adjusted or delegated by the new member themselves, implementing a form of MAC. OAuth scopes provide a way to layer DAC on top of an existing MAC security layer, or to provide finer-grained distinctions than a DAC system provides.

Putting the theoretical distinction between MAC and DAC to one side, the more practical distinction between scopes and permissions relates to how they are designed. The administrator of an API designs permissions to reflect the security goals for the system. These permissions reflect organizational policies. For example, an employee doing one job might have read and write access to all documents on a shared drive. *Permissions* should be designed based on